



Praktikum Design of Wind Farms

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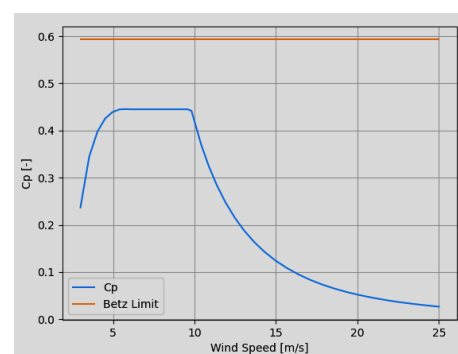
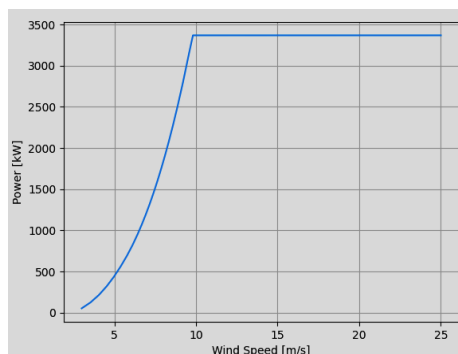
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Assignment 4 (11%): Wind Farm Cost Analysis Using FLORIS + LandBOSSE

This assignment aims at introducing students to the use of FLORIS and LandBOSSE including plotting the wind rose of a site, defining a wind farm in FLORIS, and calculating the AEP and LCOE of the wind farm. Furthermore, students learn how to work with LandBOSSE as a tool to estimate the cost of different sectors for building a wind farm.

Three IEA 3.4MW wind turbines are planned to be installed in a row (north to south) with a separation distance of 5D between each successive turbine. The substation is placed perpendicular to the turbine line, with a distance of 500m to their center point. The wind shear exponent and TI are 0.14 and 6%, respectively. The key parameters of the turbines and their characteristics are as follows:

Item	Value	Units
Name	IEA 3.4MW Reference	N/A
Rated Power	3370	kW
Rated Wind Speed	9.8	m/s
Cut-in Wind Speed	3	m/s
Cut-out Wind Speed	25	m/s
Rotor Diameter	130	m
Hub Height	120	m
Control	Pitch Regulated	N/A



A .csv file is available on [GitHub](#).

1. Create and plot the wind rose of the site based on the provide time series ("sample_time_series.csv") using Floris (hint: in Floris v4, check the example file "003_wind_data_objects.py").
2. Define the wind farm as well as the windrose in FLORIS using the Gauss wake model (default parameters). Then, determine the AEP, the capacity factor of the wind farm as well as the wind farm efficiency (P/P_{no_wake}). Use a wind rose discretization with a bin size



of 1m/s for wind speed and 1° for wind direction (Hint: check the FLORIS example file “003_wind_data_objects.py” and “006_get_farm_aep.py”).

3. Define the wind farm in LandBOSSE. Use “iea36_120_project_data” as the input turbine and update the “Turbine rating MW” as well as the rest of the parameters needed to be updated based on the following criteria. Then, calculate the Balance-of-Plant costs for the project and the LCOE.
 - The fuel cost per gal is 5.19 USD.
 - The line frequency is 50 Hz.
 - The array voltage is 30kV.
 - The turbine cost is approximately \$1.3 million per megawatt (MW).
 - The O&M cost is 0.016 \$/kWh.
 - The rental payment for land is \$20,000/MW-yr.
 - The discount rate is 4% and the design lifetime of the farm is 20 years.
 - The project construction time is 12 months.
 - The rest of the parameters in LandBOSSE remain as default. (all parameters associated with the turbine specifications should be updated)
4. Compare the estimated LCOE with the literature and plot the cost breakdown of the wind farm by turbine investment, balance-of-plant investment, O&M, and land rental in a pie chart.
5. Repeat parts 2 and 3 if the separation distance increases to 8D. Calculate the relative difference (percentage) of the AEPs and LCOEs with respect to the case of 5D.
6. If the economic situation changes in the country and the discount rate decreases by 10%, what would be the LCOE for the case of 5D separation distance? Calculate the relative difference (percentage) of the LCOEs with respect to the case of the discount rate of 4%.
7. Name (at least) one method to improve (optimize) the AEP of this installed wind farm.
8. For the case of question 3 determine the internal rate of return (IRR) numerically or graphically (hint: plot NPV for a range of discount rates) and calculate the profitability index (PI) if the average electricity spot price is 41.8 \$/MWh over the whole lifetime of the project. Is the project economically attractive? If not, explain how this project might become economically viable.
9. For the same case, estimate the minimum number of turbines in a straight North-South line with a turbine spacing of 5D to achieve $PI > 1$. Compare LCOE and PI with growing turbine number.

Note: Please upload your code, and a word / pdf file in which you describe the steps of your solution. It is not enough to only upload code files.